

Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

FACT-AI Course Project

January 28, 2026

Bilinear MLPs: Weight-Based Interpretability

Purpose:

- ▶ Direct interpretability from weights
- ▶ No post-hoc analysis needed
- ▶ No SAE training required

Structure:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x})$$

Benefits:

- ▶ Element-wise multiplication creates quadratic interactions
- ▶ Interaction tensor can be eigendecomposed
- ▶ Interpretable features directly from weights

Eigendecomposition for Interpretability

Method:

1. Extract interaction tensor for each class c :

$$B_c = \mathbf{W}_l \otimes \mathbf{W}_r$$

2. Symmetrize the tensor:

$$B_c^{\text{sym}} = \frac{1}{2}(B_c + B_c^\top)$$

3. Eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^d \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top$$

Effective Rank:

$$\text{EffRank}(\lambda) = \left(\frac{\|\lambda\|_1}{\|\lambda\|_2} \right)^2$$

What Went Well: Vision Results

MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors

Key Result:

- ▶ Effective rank: 15.5 (WD) vs 38.5 (none)
- ▶ Ratio: $0.40 < 0.5$ ✓
- ▶ Accuracy: 97–98%

Decomposition Formula:

$$B_c^{\text{sym}} = \sum_{r=1}^d \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^T$$



Figure: Regularized eigenvectors

Vision Results: Key Figures

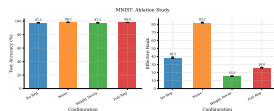


Figure: Accuracy vs Effective Rank

$$B_c^{\text{sym}} = \sum \lambda_r \mathbf{v}_r \mathbf{v}_r^T$$

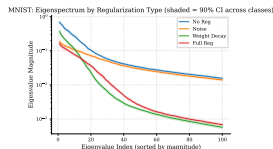


Figure: Eigenspectrum Comparison

$$\text{EffRank} = (\|\lambda\|_1 / \|\lambda\|_2)^2$$

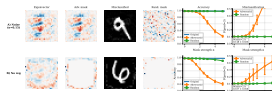


Figure: Adversarial Robustness

Eigenvector-based attacks

Language: Negation Circuit Discovery (Figure 8)

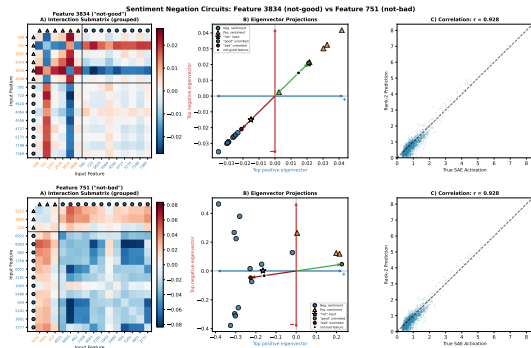
Interaction Matrix:

$$\mathbf{z}_f^{\text{out}} \approx (\mathbf{z}^{\text{in}})^{\top} \mathbf{Q}_f \mathbf{z}^{\text{in}}$$

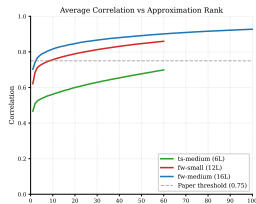
Key Insight:

- ▶ SAEs $\rightarrow$ sparse features
- ▶ Eigendecompose $\mathbf{Q}_f$
- ▶ Reveals AND-gate circuit:

negation $\otimes$ sentiment
 $\downarrow$
 negated-sentiment

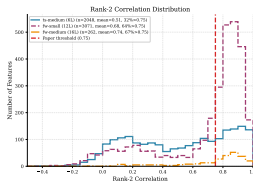


Language: Low-Rank Correlation Analysis (Figure 9)



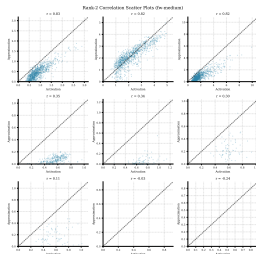
Rank- k Approximation:

$$\mathbf{z}_f^{\text{out}} = \sum_{r=1}^k \lambda_r (\mathbf{v}_r^{\top} \mathbf{z}^{\text{in}})^2$$



Results:

- Paper: 69% features >0.75 at rank-2
- Ours: 18–32% (SAE training gap)



Reproduction Summary

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	✓	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	✓	Clear digit patterns with regularization
V3	Overfitting without reg.	✓	Diffuse patterns without regularization
V4	Regularization roles	✓	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	✓	Std < 1% across 5 seeds
V6	Adversarial generation	✓	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	✓	Pretrained SAEs work for analysis
L2	Negation/AND-gate circuits	✓	Opposing eigenvalues confirmed
L3	Low-rank interactions	~	18–32% >0.75 vs. paper's 69%
L4	Sparse interactions	✓	Clear block structure in feature 751
L5	Cross-model generalization	~	Methodology works; magnitudes differ

Extension 1: Background and Purpose

Hypothesis: Eigenvector decomposition generalizes across datasets

Test Cases:

- ▶ **Writer Independence:** MNIST $\leftrightarrow$ EMNIST-Digits
- ▶ **Domain Transfer:** MNIST $\rightarrow$ USPS ($16 \times 16 \rightarrow 28 \times 28$)
- ▶ **Geometric Semantics:** MNIST $\rightarrow$ EMNIST-Letters
 - ▶ $O \rightarrow 0, I \rightarrow 1, Z \rightarrow 2, S \rightarrow 5$

Intuition: Geometric resemblance should lead to similar eigenvectors

All models use Center-of-Mass (CoM) normalization to focus on shape geometry

Extension 1: Results - Generalization Works!

Cross-Dataset Accuracy:

- ▶ MNIST $\rightarrow$ EMNIST-Digits: **97.7%**
- ▶ EMNIST-Digits $\rightarrow$ MNIST: **97.9%**

USPS Transfer:

- ▶ Regularized: **72.3%**
- ▶ Baseline: 53.4%
- ▶ Improvement: **+19 pp**

Letter $\rightarrow$ Digit Mapping:

- ▶ O $\rightarrow$ 0: **99.6%**
- ▶ I $\rightarrow$ 1: **87.1%**
- ▶ Z $\rightarrow$ 2: **88.8%**
- ▶ S $\rightarrow$ 5: **87.4%**

Extension 1: Cosine Similarity Limitations

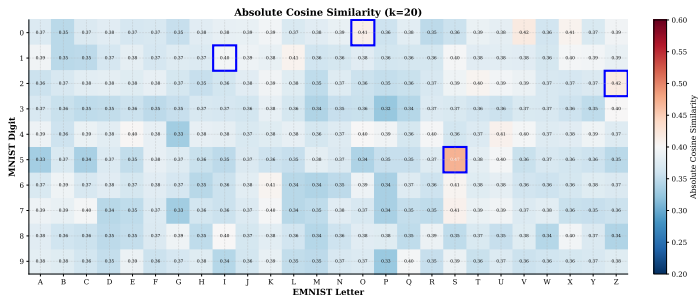


Figure: Absolute cosine similarity between MNIST digit classes and EMNIST letter classes fails to distinguish geometrically similar pairs (0–O) from dissimilar pairs (0–X).

Cosine similarity fails: Similar pairs (0–O): 0.358 vs. Dissimilar pairs (0–X): 0.339 (virtually identical)

Quadratic Form Similarity

Problem: Cosine similarity fails

- ▶ Similar pairs (0–0): 0.358
- ▶ Dissimilar pairs (0–X): 0.339
- ▶ Not discriminative!

Solution: Quadratic Form Similarity

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1]$$

Results:

- ▶ Similar pairs: **0.400**
- ▶ Dissimilar pairs: **-0.060**
- ▶ Large gap with $p < 10^{-4}$

Incorporates eigenvalue signs and eigenvector alignment

Extension 1: Quadratic Form Similarity Heatmap

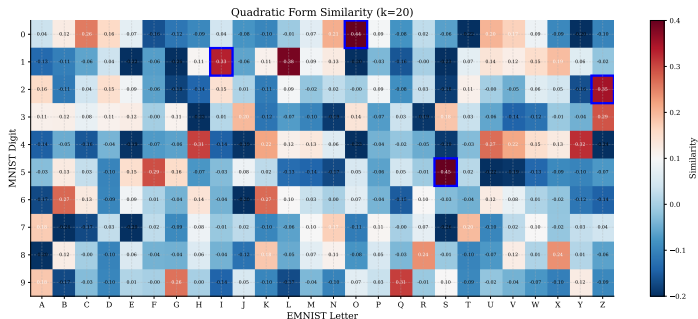


Figure: Quadratic Form Similarity Heatmap

Clear discrimination: strong diagonal for MNIST vs. EMNIST-Digits; bright entries at shape-matched pairs (0-O, 1-I, 2-Z, 5-S)

Extension 1: Key Figures

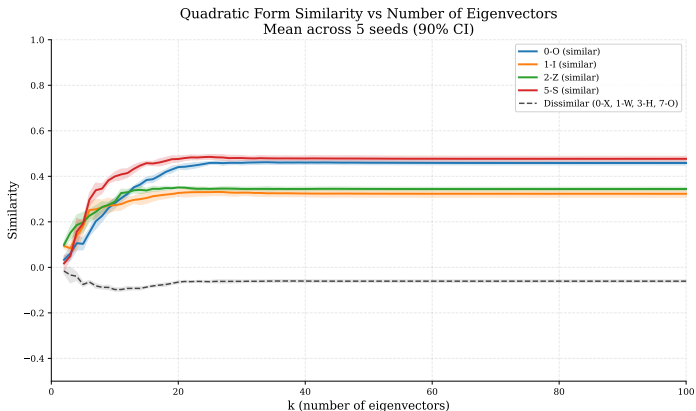


Figure: Quadratic Form Similarity vs number of eigenvectors k . Similar pairs converge around $k = 20$; dissimilar pairs stay near zero.

Extension 2: CP-Decomposition

Motivation:

- ▶ Post-hoc → **intrinsic** interpretability
- ▶ Standard eigendecomposition: orthogonality → entangled
- ▶ CP: disentangled features

CP Factorization:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr}$$

Key Properties:

- ▶ No orthogonality constraint
- ▶ Disentangled representations
- ▶ Parameter reduction → faster training and scalability

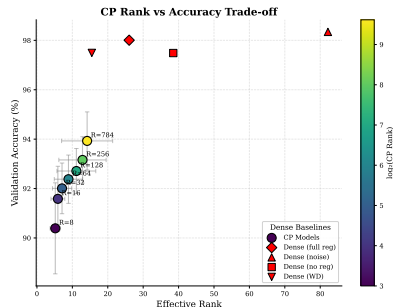


Figure: Accuracy vs rank trade-off

Results at $R = 256$:

- ▶ Accuracy: **93.8%**
- ▶ Effective rank: **17.5**
- ▶ vs Dense: 97.5% (3.7 pp drop)

CP-Decomposition: Efficiency

Experiment	Runs	Wall (h)	GPU-h	CO2 (kg)
Vision (Section 4) [†]	81	1.22	1.11	0.039
Language (Section 5) [‡]	2	0.19	0.07	0.005
Extension 1	15	0.58	0.58	0.020
Extension 2 (CP)	105	0.05	0.05	0.004
Total	203	2.04	1.81	0.067

Table: Environmental impact (CO2 via codecarbon)

Key Efficiency Gains:

- ▶ **Parameter reduction:** $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2) \rightarrow \mathcal{O}(R \times d_{\text{in}})$
- ▶ Average GPU-h per run: CP decomposition (0.00048) vs Vision (0.014) — **29× more efficient**
- ▶ Only **3.7 pp accuracy drop** for significant parameter efficiency

CP-Decomposition: Disentangled Factors

Dense Baseline (Full Reg)

Top 5 Eigenvectors Comparison: Dense vs CP

CP Model (R=64, lambda)



CP-Decomposition Variants

Three variants with different regularization approaches

Fixed CP

Fixed CP: Hard rank constraint with no additional regularization

Characteristics:

- ▶ Strict complexity bound: rank R is fixed during training
- ▶ No regularization on the factors or scaling parameters
- ▶ Operates as a strict bottleneck

Results:

- ▶ Effective rank ~ 3.5 regardless of allocated capacity ($R \in \{32, 64, 128, 256\}$)
- ▶ Accuracy: $\sim 91.9\%$ (vs. 97.5% dense)
- ▶ $\sim 6\%$ accuracy cost for extreme sparsity

Forces extreme sparsity at the cost of accuracy

Lambda CP (L1 Regularization)

Lambda CP: L_1 regularization on learnable scales λ_r

Characteristics:

- ▶ Soft sparsity via L_1 regularization on λ_r parameters
- ▶ Allows factors to be pruned naturally during training
- ▶ More flexible than Fixed CP

Results:

- ▶ At $R = 256$: **93.8%** accuracy
- ▶ Effective rank: **17.5**
- ▶ Comparable to dense models with weight decay (eff. rank 15.5)

Provides robust middle ground: competitive accuracy with structural disentanglement

Gated CP (L0 Proxy)

Gated CP: Hard sigmoid gates with L_0 proxy for explicit rank pruning

Characteristics:

- ▶ Uses L_0 regularization proxy: $\sum_r \sigma(\phi_r)$ approximates count of active ranks
- ▶ Learnable gate logits ϕ_r control continuous gate values (0–1)
- ▶ Hard sigmoid gates: $\text{clamp}(\sigma(\phi_r) \cdot 1.1 - 0.05, 0, 1)$
- ▶ Most flexible regularization approach

Results:

- ▶ At $R = 256$: **93.79%** accuracy
- ▶ Effective rank: **17.35**
- ▶ Similar performance to Lambda CP

Alternative approach to Lambda CP with similar performance; both provide efficient, interpretable modeling